Jiming Wang

Last updated: October 22, 2026

Department of Economics, The Pennsylvania State University
Kern Building, University Park, PA 16802, USA

jimwang@psu.edu
+1 (814) 699-0030

Education

2021 – 2027 (Expected)	Ph.D. in Economics, *Penn State, USA*
2018 – 2021	M.A. (Doctorate Courseworks) in Economics, *Peking University, China*
2014 – 2018	B.S. in Mathematics and Physics, *Tsinghua University, China*
2015 – 2018	B.A. (Double Major) in Economics, *Peking University, China*
Spring 2017	Visiting Undergraduate Student, *National University of Singapore*

Fields of Interest

Industrial Organization, Environmental Economics, Computational Economics, Applied Microeconomics

References

Mark J. Roberts
Professor Emeritus of Economics
The Pennsylvania State University
mroberts@psu.edu

Andres Aradillas-Lopez
Professor of Economics
The Pennsylvania State University
aza12@psu.edu

Paul L. E. Grieco
Professor of Economics
The Pennsylvania State University
plg15@psu.edu

Kenneth L. Judd
Senior Fellow
Hoover Institution, Stanford University
kennethjudd@mac.com

Working Papers

Spatial Heterogeneity of Externalities in EV Charging Investments

(Job Market Paper, with Jianwei Xing)

Private investment in EV charging infrastructure can be spatially misallocated when externalities and costs vary within cities. Using data on public charging stations, EV sales, and real-time station utilization in China, we estimate a dynamic investment game between charging service providers. We distinguish high-density urban cores from low-density peripheries and quantify two opposing externalities. The indirect network effect of charging capacity on EV purchases is roughly twice as large in the core as in the periphery, leading to underinvestment in the core. By contrast, the business stealing effect between providers is about four times as large in the periphery, inducing overinvestment there. Using the estimated model, we find that private investment is about 22% below the social optimum in the core and 9% above it in the periphery. These distortions generate an annual welfare loss of about 3.87 million RMB for a representative city. Maximizing city surplus requires subsidizing new capacity in the core and taxing it in the periphery. Our results imply that accounting for spatial heterogeneity is essential for effective policy design.

Heterogeneity of Indirect Network Effects: Empirical Analysis of the Chinese Electric Vehicle Market

(with Jianwei Xing, Junji Xiao, and Xiaodan Guo)

Revise and Resubmit, Journal of the Association of Environmental and Resource Economists

Governments can accelerate technology adoption by directly subsidizing the technology or supporting complementary infrastructure, particularly when indirect network effects are present. This paper examines the heterogeneity of these indirect network effects, driven by variations in product characteristics within the market. Using data on electric vehicle (EV) sales and charging infrastructure provision in China, and focusing on a period of rapid market expansion with increasing product differentiation, we analyze the indirect network effects between EV adoption and charging stations, investigating their heterogeneity. We also assess the relative effectiveness of EV purchase subsidies and charger subsidies in promoting EV adoption. Our findings reveal that the strength of indirect network effects depends on vehicle characteristics, particularly driving range. Furthermore, the two subsidy instruments have distinct distributional effects: purchase subsidies primarily benefit high-range vehicles and their buyers, while charger subsidies disproportionately favor low-range vehicles and their buyers.

Growth or Redistribution? The Business Effects of EV Charging Infrastructure in Kansas City
(with Ruohao Zhang, Xibo Wan, Zhen Lei, and Junxian Tang)
Under Review
Public infrastructure such as EV charging stations (EVCS) can raise nearby business activity, but these gains may reflect redistribution rather than net growth. Using cellphone mobility data on restaurant visits in Kansas City, we estimate a random-coefficient logit model. To address endogenous EVCS placement, we develop novel spatial instruments using algorithmically simulated charging station locations based on citywide road geography and future deployment in peripheral rings. Our estimates show that consumers value a nearby EVCS at $5.23 per restaurant visit, equivalent to 11 extra minutes of one-way travel. While EVCS generate a significant gain in restaurant visits, about three-quarters of the gain reflects redistribution, suggesting that evaluations based on local business gains can overstate the economic benefits of infrastructure investment. Redistribution also matters for place-based policy. Counterfactual simulations show that targeting EVCS to disadvantaged tracts within the city makes restaurant visits more unevenly distributed in those communities.

Work in Progress

Rule-Bound Buyers Seek Discretion Most? Procurement Discretion, Thresholds, and Supplier Selection in China (with Songrui Liu, Guangrong Ma, and Yaohong Zhao)

Presentations

2027	Econometric Society NA Winter Meeting (ASSA, DC; scheduled)
2026	Trade, Development, & Macro Brownbag (Penn State); IO Brownbag (Penn State); Camp Resources XXXII (Asheville); European Association for Research in Industrial Economics Annual Conference (Rising Star Session; Mannheim, Germany); Urban Economics Association NA Meeting (Chicago); Economics of EV Charging Workshop (Harvard; scheduled)
2022–2025	Chinese Economists Society NA Annual Conference (Lewisburg; 2024); CCER Summer Institute (Chongli, China; 2023); IO Brownbag (Penn State)

Teaching Experience

The Pennsylvania State University (as Instructor)

Summer 2025 & 2026	ECON 104: Introductory Macroeconomic Analysis and Policy (online)
Summer 2023	ECON 304: Intermediate Macroeconomic Analysis (in person)

The Pennsylvania State University (as Teaching Assistant)

Fall 2026	ECON 454: Economics of Mergers

Peking University (as Teaching Assistant)

Spring 2020 & 2021	International Trade (undergraduate)
Fall 2019	Advanced Macroeconomics I (Ph.D. Level)

Research Assistant Experience

2025 – 2026	RA for Prof. James Tybout, *Penn State*
Summer 2022	RA for Prof. Mark Roberts, *Penn State*
Before 2021	RA for Prof. Jianwei Xing, *Peking*; RA for Prof. Alexander White, *Tsinghua*

Fellowships and Awards

2026	Ed Green Graduate Fund, *Penn State*; UPAC Travel Fund, *Penn State*
2025	RGSO Dissertation Support, *Penn State*; UPAC Travel Fund, *Penn State*
2021 – 2026	Graduate Economics Scholarship, *Penn State*
Before 2021	Excellent Graduate, *Peking*; China's National Scholarship, *Peking*

Skills

Programming:	Julia, Python, MATLAB, Stata, C, Mathematica, AMPL, QGIS, LaTeX
Languages:	English (fluent), Chinese (native)